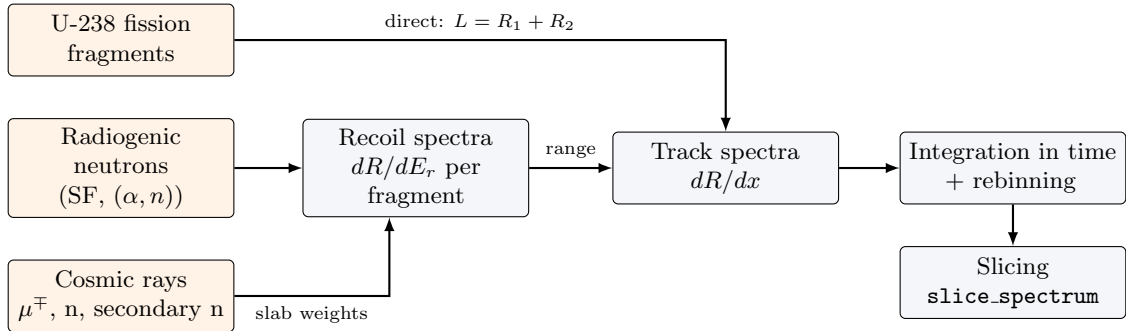


October 2026

Overview

mineral_utils turns Geant4 output into expected track-length spectra in a mineral. Three families of Geant4 runs are used:

- **mineral runs** (Geant4_data/<mineral>/<species>/outNuclei<E>.txt): the population of recoils produced per primary of energy E (the target size only provides statistics);
- **StdRock runs** (Geant4_data/StdRock<species>/): where along the path the primaries stop or interact, used for any overburden and slab position;
- **range tables** (Geant4_data/<mineral>/Range/DEDX_Z<Z>_A<A>.txt): stopping power and CSDA range of every nucleus in the mineral.



1 Main constants

Constant	Value	Meaning
PROTON_MASS_MEV, NEUTRON_MASS_MEV	938.3, 939.6 MeV	Nucleon masses for fragment masses
BO_U238	7.570126 MeV	Binding energy per nucleon of U-238
FISSIONS_U238_G_KYR	2.14×10^8	Spontaneous fissions per g of U-238 per kyr
U238_ABUNDANCE	0.9927	U-238 fraction of natural uranium
KYR_PER_SECOND	$1/3.1536 \times 10^{10}$	Converts per-second rates to per-kyr
NU_BAR_U238_SF	2.0	Neutrons per spontaneous fission
WATT_A_MEV, WATT_B_PER_MEV	0.65 MeV, 3.7 MeV^{-1}	Watt spectrum of the SF neutrons
RECOIL_ENERGY_BINS_MEV	301 log edges, 10^{-3} – 10^4 MeV	Recoil-energy binning
TRACK_LENGTH_BINS_NM	1000 log edges, $10^{1.5}$ – $10^{5.5}$ nm	Fine track-length grid for the calculation
DEFAULT_INTEGRATION_X_BINS_NM	200 edges, 0–50 000 nm	Default output binning (settable)
_GEANT4_ENERGY_BINS_GEV_FULL	<code>np.logspace(-3, 4, 120)</code>	Simulated primary energies; must match the file names
GEANT4_MAX_SIMULATED_ENERGY_GEV	196.84 GeV	Upper edge: 91 edges, 90 bins; bin i uses the run at its lower edge
STOPPING_N_QUANTILES	100 (+ 8 tail nodes)	Quantile nodes of the StdRock depth distributions
MIN_DEPTH_SAMPLES	20	Energies with fewer depths are skipped
DEPTH_FLOOR_MWE	10^{-6} m.w.e.	Floor for log interpolation
TRUNCATION_FRACTION	0.98	Depths reaching 98% of the StdRock target: energy skipped
CAPTURE_SPECTRUM_MAX_E0_GEV	1 GeV	μ^- runs used for the capture spectrum
NEUTRON_TAIL_P	0.95	Neutron CDF: empirical below, exponential tail above
slab	1 mg cube	$t = (10^{-3}/\rho)^{1/3}$ cm, $t_{\text{mwe}} = 0.01 t \rho$
total_simulated_particles	10^4	Primaries per Geant4 run
integration grids	50×50 , $\cos \theta \in [0.01, 1]$	Energy $\times$ zenith grid inside each energy bin

2 Common step: recoil energy to track length

For each fragment (Z, A) the range table gives $E(x)$ (inverse CSDA range) and the total stopping power $S(x) = dE_e/dx + dE_n/dx$. Light ions are mapped to H1, H2, H3, He4.

Key passage — `_convert_recoil_to_track_spectrum`

$$\frac{dR}{dx}(x) = \frac{dR}{dE_r}(E(x)) S(x) \times 10^{-3} \quad [\text{tracks mg}^{-1} \text{ kyr}^{-1} \text{ nm}^{-1}],$$

summed over all fragments (neutrons excluded). The factor 10^{-3} converts per g to per mg.

The counts over an exposure are $N(x) = \int dt dR/dx$ (trapezoidal rule on `steps` times), then rebinned from the fine grid to `x.bins` through the cumulative distribution.

3 Spontaneous fission

`calculate_fission_spectrum` uses the fragment pairs in `U238.dat` and the binding energies in `BindingEne.txt`.

Key passage — fragment energies and track length

$$M_0 = 92m_p + 146m_n - 238B_0, \quad m_k = Z_k m_p + (A_k - Z_k)m_n - A_k B_k,$$

$$E_1 = \frac{M_0^2 + m_1^2 - m_2^2}{2M_0} - m_1, \quad E_2 = \frac{M_0^2 + m_2^2 - m_1^2}{2M_0} - m_2, \quad L = R_1(E_1) + R_2(E_2).$$

$$\frac{dR}{dx} = \frac{n(L \in \text{bin})}{N_{\text{used}}} C_U F_{238} a_{238} \frac{1}{\Delta x} \quad [\text{tracks g}^{-1} \text{ kyr}^{-1} \text{ nm}^{-1}],$$

with C_U the uranium concentration (g/g), $F_{238} = \text{FISSIONS.U238.G.KYR}$, $a_{238} = \text{U238.ABUNDANCE}$. Events with a fragment missing from `BindingEne.txt` are skipped and not counted in N_{used} .

`integrate_fission_spectrum` multiplies by the age and by 10^{-3} (per mg). No Geant4 recoil step is involved: each fission gives one track whose length is the sum of the two fragment ranges.

4 Radiogenic neutrons

Two sources, both proportional to the uranium concentration. The SF neutrons share the fission rate of the fragment tracks, $R_f = C_U a_{238} F_{238}$ (`_fission_rate_per_g_kyr`):

- **spontaneous fission** (`_radiogenic_spectrum`): Watt shape $e^{-E/a} \sinh \sqrt{bE}$ with $a = 0.65$ MeV, $b = 3.7$ MeV $^{-1}$, normalised to $R_f \bar{\nu}$ ($\approx C_U \times 0.0135$ n g $^{-1}$ s $^{-1}$ for $\bar{\nu} = 2$);
- (α, n) (`_alpha_n_spectrum`): shape $E^2 e^{-\beta E}$ ($\beta = 900$ GeV $^{-1}$) with a smooth cut-off at 6.5 MeV, normalised to $C_U \times k_{\alpha n} \times 8 \times 1.245 \times 10^4 \times 10^{-6}$ n g $^{-1}$ s $^{-1}$ (8 α per U-238 chain, 1.245×10^4 Bq/g U, $k_{\alpha n} = \text{an_coeff_n_alpha}$ in 10^{-6} n/ α).

Key passage — `_process_background_neutron_geant4_data`

The neutrons are born inside the mineral, so every recoil they produce counts (infinite homogeneous medium):

$$\frac{dR}{dE_r} = \sum_i \left[\int_{E_i}^{E_{i+1}} S_n(E) dE \right] \frac{n_i(E_r)}{N \Delta E_r} \cdot \frac{1}{\text{KYR.PER.SECOND}} \quad [\text{recoils g}^{-1} \text{ kyr}^{-1} \text{ MeV}^{-1}],$$

with S_n the emission spectrum and n_i the recoil histogram of the neutron run at E_i . No depth or slab weighting enters.

5 Signal particles: μ^- , μ^+ , cosmic neutrons

Slab. The sample is a 1 mg cube under an overburden $X(t)$ [m.w.e.] given by the overburden history. Along a direction with $c = \cos \theta$, the slab spans $z_{\min} = X/c$ to $z_{\max} = (X + t_{\text{mwe}})/c$.

Depth models from StdRock. For each simulated energy the depths are stored as quantile nodes and interpolated in log-log in energy; F is piecewise linear between nodes.

- muons: stopping depth R (rows with remaining primary energy 0), with $G(z) = \int_0^z (1-F) = E[\min(R, z)]$;
- neutrons: all interaction depths, empirical up to the 95% quantile and an exponential tail beyond it (length = mean excess depth), so deep slabs keep a finite probability.

Key passage — slab probabilities

$$P_{\text{int}}^{\mu} = \frac{G(z_{\text{max}}) - G(z_{\text{min}})}{E[R]}, \quad P_{\text{stop}}^{\mu} = F_R(z_{\text{max}}) - F_R(z_{\text{min}}), \quad P^n = F_n(z_{\text{max}}) - F_n(z_{\text{min}}).$$

The muon interaction density is proportional to $P(R > z)$, the fraction of muons still moving.

Key passage — `_process_geant4_data`

For each primary-energy bin i and time t :

$$w_i(t) = \int_{E_i}^{E_{i+1}} dE \int_{0.01}^1 dc \Phi(E, t) P(z_{\text{min}}, z_{\text{max}} | E),$$

$$\frac{dR}{dE_r}(t) = \frac{\sum_i w_i(t) n_i(E_r)}{\Delta E_r t \rho N \text{ KYR_PER_SECOND}} \quad [\text{recoils g}^{-1} \text{ kyr}^{-1} \text{ MeV}^{-1}].$$

n_i is the recoil histogram of the mineral run at the lower edge E_i ; dividing by $t\rho$ (g/cm² of the slab) gives a rate per gram. For μ^- : n_i excludes captures (remaining energy > 0) and is weighted with P_{int}^{μ} ; captures add $[\sum_i w_i^{\text{stop}}(t)] \bar{n}_{\text{cap}}(E_r)$, where w^{stop} uses P_{stop}^{μ} and $\bar{n}_{\text{cap}}$ is the mean capture histogram of the runs with $E_0 \leq 1$ GeV (all muons stop there, so it is per stopped muon). μ^+ uses only P_{int}^{μ} with the μ^- table; neutrons use P^n .

The flux is evaluated on the FluxHistory timeline, $t_{\text{flux}} = t - t_{\text{age}}$ (0 = present).

Secondary neutrons (`species='secondary_neutron'`). Like the primaries, they are transported through the overburden with the StdRock runs, and only the final recoils come from the mineral. For a direction c the path to the sample has slant depth $S = X(t)/c$; only neutrons born in the window $[S - W, S]$ matter, where W is the depth at which a neutron survives with probability 10^{-4} (from the StdRock neutron table), split into 50 bins of width $\delta = W/50$.

Key passage — source and generations (`_secondary_neutron_yield`)

$$A_{b,m} = \sum_s \sum_i \Phi_{s,i}(t) \frac{n_i^{\text{StdRock}_s}(b, m)}{N}, \quad Y_b \left(\mathbb{K} - \frac{1}{2} C_n[0] \right) = A_b + \sum_{b' < b} Y_{b'} \frac{1}{2} (C_n[b - b' - 1] + C_n[b - b']),$$

$n_i(b, m)$: neutrons of energy cell m produced in window bin b by the StdRock run at E_i , at their actual slant depth, so the transport of the primary down to the window (energy loss, stopping) is included. $C_n[d]$: neutrons produced by a neutron at offset bin d , from the StdRock neutron runs; neutrons are born uniformly within their bin. μ^+ uses StdRock_mu+ if present, otherwise the μ^- runs without the stopping step (no capture neutrons).

Key passage — propagation to the slab (`_process_secondary_geant4_data`)

Neutrons keep the primary's direction. For a neutron born uniformly in bin b , at distance $d \in [d_0, d_0 + \delta]$ from the slab (path $\tau = t_{\text{mwe}}/c$ in the slab), with $H(z) = \int_0^z F_n = z - G(z)$:

$$\bar{P}_b = \frac{H(d_0 + \delta + \tau) - H(d_0 + \tau) - H(d_0 + \delta) + H(d_0)}{\delta}, \quad w_m(t) = \int_{0.01}^1 dc \sum_b Y_{b,m}(t, c) \bar{P}_b(E_m, c).$$

The recoils per neutron of the mineral run at E_m are applied with the normalisation of `_process_geant4_data`.

Limits: depths are bounded by the StdRock target length (as for the muon tables; a warning is printed beyond it); secondary neutrons keep the primary direction; neutrons below 1 MeV are not followed; neutrons born above the window are neglected ($< 10^{-4}$ survival).